

Tan Dai Ngo

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EDUCATION

University of Victoria | Master of Engineering, Applied Data Science *Victoria, BC | Sep 2025 - Dec 2027*

- **Coursework:** Optimization for ML, Artificial Intelligence, Data Mining. **GPA:** 8.78 / 9.00.

University of Chicago | Master of Science, Computer Science *Chicago, IL, USA | Sep 2020 - Mar 2022*

- **Coursework:** Algorithms, Distributed Systems, Databases, Applied Software Engineering, Advanced Python Programming.

University of Washington | Bachelor of Science, Economics *Seattle, WA, USA | Sep 2015 - Aug 2019*

- **Minors:** Informatics, Applied Mathematics. **GPA:** 3.57 / 4.00.

EXPERIENCE

Software Developer | T-Mobile *Bellevue, WA, USA | Jun 2022 - Apr 2024*

- Built REST endpoints across 5+ roaming microservices in Java and Spring Boot, holding queries under 3 seconds.
- Cut roughly 80% of a recurring weekly workload by automating rejected test reprocessing with Kafka driven retries.
- Traced 20+ cross service defects to root cause from Splunk logs on live production traffic.
- Shipped 50+ releases through a Jenkins CI/CD pipeline, reviewing teammates' changes in Git.
- Delivered a test rejection notifier over the Microsoft Graph API so failures reached the team directly.

Math and Computer Science Teacher | APU American International Schools *Da Nang, Vietnam | Aug - Dec 2024*

- Taught Python, algorithms and software engineering fundamentals to 50+ students through hands on projects.

PROJECTS

Multi-Stage IoT Intrusion Detection | Python, PyTorch, XGBoost, graph neural networks *Capstone, team of 7, UVic*

- Benchmarked six detection models to 99.6% recall at 0.82% false positives under a fixed 1% false alarm budget.
- Designed the four condition holdout that exposed session leakage in the benchmark, then re-scored every model on an honest split where PR-AUC landed at 0.242.
- Scripted an integrity check confirming all 23 engineered features stayed causal and label free.

Ocean Data ML Platform with RAG | Python, Spark, XGBoost, Sentence Transformers, Qdrant, FastAPI *Personal Project*

- Ingested 10M+ NOAA and Ocean Networks Canada observations into raw, cleaned and curated Spark layers, validating each tier so nothing downstream read partial data.
- Served plain English retrieval over that corpus at hit@k 0.90 and term recall 0.85, using embeddings, a vector store and a local model behind FastAPI.
- Validated the forecasting pipeline on air temperature, cutting RMSE 48% at the 12 hour horizon.

Distributed Facility Reservation System | Python, REST, OpenAPI, SQLite, pytest *Team of 4, UChicago*

- Designed 27 versioned REST endpoints published as OpenAPI documentation for four other teams.
- Wrote 71 automated tests across the API, database and rules layers as the release gate.
- Secured every endpoint behind session tokens checked for freshness and permission scope.

TECHNICAL SKILLS

Languages: Python, Java, Go, SQL, JavaScript, Bash

Machine Learning & AI: PyTorch, XGBoost, scikit-learn, DistilBERT, Optuna, MLflow, Fairlearn, RAG, Ollama

Data & Backend: Spark, PySpark, Kafka, Cassandra, MySQL, MongoDB, Qdrant, FastAPI, Spring Boot, REST

Platform & Tools: Docker, AWS (EC2, S3), Jenkins CI/CD, Git, Linux, Splunk, pytest, Jira, Confluence